

K SHREEKRISHNA UPADHYAYA

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PROFILE

Curious engineering student with an interest in low-level and system development. Learns primarily through experimentation and building practical, hands-on projects.

EDUCATION

VIVEKANANDA COLLEGE OF ENGINEERING AND TECHNOLOGY, PUTTUR

Bachelor of Engineering in Artificial Intelligence & Machine Learning

Cumulative GPA (till 5th semester): 8.8/10.0

Relevant Coursework: Data Structures, OS, Computer Networks, Automata Theory

DK, Karnataka

2023 – Present

ACTIVITIES & CONTRIBUTIONS

Hash Shell | C

Contributor

- Optimized interactive shell behavior by resolving multi-line continuation bugs in quoted strings.
- Fixed tab-completion prefix errors and ensured consistent terminal state during interactive mode.
- Refactored long functions and aligned code with the coding standards according to contributing guidelines.
- Adhered to upstream coding standards and performance requirements during the PR review process.

PROJECTS

Sn-prefix Libraries | C, x86_64 ASM

Present

- Developing a collection of cross-platform libraries for reuse across multiple projects with minimal dependencies.
- Implemented cross-platform file system access, virtual memory, and synchronization tools.
- Built synchronous and asynchronous logging and tracing utilities.

Regex Engine | C

December 2025

- Built a parser for regex patterns supporting basic operators (wildcards, quantifiers, concatenation, grouping).
- Implemented Thompson-style NFA construction and simulation for pattern matching in input text.

StateFlow: DFA and NFA simulator | C, Raylib

September 2025

- Built a visual simulator to show state transitions for DFA and NFA automata.
- Created a UI to design custom automata and validate formal logic through step-by-step simulation.

Cross-Platform Game Engine Core | C, x86_64 ASM, Vulkan

February – May 2025

- Implemented atomic operations using inline and MASM x86_64 assembly for thread-safety.
- Built a thread abstraction layer and manual memory management using custom allocators.
- Developed a windowing abstraction layer supporting Win32, X11, and Wayland backends.
- Implemented input and event subsystems.

LC3VM & Kilo Editor | C

2024 – 2025

- Implemented the LC-3 virtual machine including instruction decoding, memory model, and execution loop.
- Built a terminal-based text editor using ncurses for buffer and cursor management.

Other Projects

- **GoAuth:** Implementation of traditional REST-based authentication service in Go.
- **WOL-Android:** Android app that runs an HTTP server and broadcasts Wake-on-LAN packets to LAN on request.
- **WallhavenAPI:** Wallhaven API client library in C and CLI tool to install wallpapers(whcli).
- **WSO:** A shell script to save and open shortcuts to web pages.

SKILLS & OTHERS

Technical Skills: C, C++, x86_64 Assembly (inline/MASM), Python, Java, Go, POSIX Shell Script; POSIX APIs, Memory Management, Threading, Synchronization; Git, Linux/Unix Shell, CMake, Typst, Vim;

Awards: Winner of **Nexathon 2025** (SDIT) and **DevHack 2025** (Sahyadri College).